

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2023-24

Paper No. BPT-401: Electronics and Modern Physics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. Answer in brief any **five** of the following:

[5 × 4 = 20]

- Show that $r^n \vec{r}$ is an irrotational vector for any value of n , but is solenoidal only if $n = -3$ (where $\vec{r}$ is the position vector of a point).
- Consider a cylindrically shaped section of silicon having length 1 mm and cross-sectional area 0.1 mm^2 . Calculate its conductivity and resistance when: (i) it is purely intrinsic material, and (ii) it has a free electron density of $8 \times 10^{15} \text{ cm}^{-3}$. (Take the intrinsic carrier density $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$, mobility constants $\mu_n = 1500 \text{ cm}^2/\text{V}\cdot\text{s}$).
- Why are transistors in common-emitter configuration preferred over other configurations in amplifier circuits? Explain.
- Describe the three electric vectors $\vec{E}$, $\vec{P}$, and $\vec{D}$. Find the relationship among them.
- An electron has a de Broglie wavelength of $0.02 \text{ \AA}$. Find its kinetic energy, phase velocity, and group velocity. (Given rest mass energy of electron = 511 keV).
- Draw the circuit diagram to obtain a clipped signal through: (i) a series diode clipping circuit, and (ii) a shunt diode clipping circuit.
- Electrons are accelerated by 344 V and are reflected by a crystal. If the first maximum is observed at 60° , calculate the interplanar spacing of the crystal. ($h = 6.62 \times 10^{-34} \text{ J}\cdot\text{s}$)

Question 2.

- Describe the electric fields in the spherical cavity of a dielectric. Derive the expression of the molecular field inside the spherical cavity and find the relation for the dielectric constant in terms of the polarizability. Give the Lorentz-Lorenz equation also. [8]
- If a dielectric is introduced between the plates of a parallel plate capacitor, show that the induced charge σ' varies with dielectric constant k as:

$$\sigma' = \sigma \left(1 - \frac{1}{k} \right)$$

Also, show that $k \rightarrow \infty$ for a metal dielectric. Here, σ is the free charge density. [6]

OR

- Distinguish among electronic, ionic, and orientational polarizability. Explain their temperature dependence. [8]

- (b) A dielectric material has relative permittivity $\epsilon_r = 6$. Calculate the electric susceptibility χ_e and the polarization $\vec{P}$ when subjected to an electric field of 10^4 V/m. [6]
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Question 3.

- (a) If the barrier potential V_0 across a p-n junction diode in terms of donor and acceptor impurities is given as:

$$V_0 = \frac{kT}{e} \ln \left(\frac{N_d N_a}{n_i^2} \right)$$

where all the symbols have their usual meanings, show that the width of the depletion region decreases with the increase of these impurities. [8]

- (b) Explain the working of a center-tapped full-wave rectifier. How can its output be smoothened? Explain with a necessary filter circuit. [6]

OR

- (a) What is a Zener diode? Explain the working of a Zener diode as a voltage regulator with a circuit diagram. [8]
- (b) Compare half-wave and full-wave rectifiers in terms of efficiency, ripple factor, and peak inverse voltage. [6]
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Question 4.

- (a) Describe the photoelectric effect. Give an account of Einstein's explanation of the photoelectric effect. The photoelectric threshold wavelength for a metal is $3000 \text{ \AA}$. Find the kinetic energy of the electron ejected from it by radiation of wavelength $1200 \text{ \AA}$. [8]
- (b) What is the Compton effect? Obtain an expression for the shift in wavelength of an X-ray beam. [6]
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Question 5.

- (a) Draw the circuit diagram to study the output characteristics of a PNP transistor in common-emitter mode. Explain the transistor operation in saturation, active, and cut-off regions. [5]
- (b) What do you understand by bias stabilization in transistor circuits? Derive the expression for the stability factor S in terms of current gain β for a CE amplifier. [5]
- (c) What is a load line? How can one determine the quiescent point (Q-point) using a load line in a CE amplifier? Why is it preferable to choose the operating point in the middle of the active region? [4]

OR

- (a) Compute the value of R_B in the biasing circuit given below so that the Q-point is fixed at $I_C = 8 \text{ mA}$ and $V_{CE} = 3 \text{ V}$. Assume $V_{BE} = 0.3 \text{ V}$, $V_{CC} = 9 \text{ V}$, $R_C = 250 \Omega$, and $\beta = 80$. [5]
- (b) Derive the Langevin-Debye equation for the total polarizability of a dielectric. [4]
- (c) State Gauss's divergence theorem. Verify the divergence theorem for the vector field $\vec{V} = x^2\vec{i} + y^2\vec{j} + z^2\vec{k}$ taken over the unit cube $0 \leq x, y, z \leq 1$. [5]